



Thermofluid Mechanics

Lab Report Portfolio

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Engine Dyno Lab Report

Abstract

This report is focused on investigating the performance of a DW12BTED4 four-cylinder diesel engine using a dynamometer under full open throttle conditions. Fuel consumption and torque were recorded against predetermined engine speeds which allowed for Brake Mean Effective Pressure (BMEP), Break Specific Fuel Consumption (BSFC) and the brake thermal efficiency to be calculated. Conclusion from the results is the power linearly increased along with engine speed up to 1750rpm with the minimum BSFC and thermal efficiency sitting in the engine speed region of 1750-2000rpm.

Introduction

The purpose of internal combustion engines (ICE's) is to convert the chemical energy into mechanical power. The working fluids inside the combustion chamber is the fuel air mixture before and after combustion. The ICE that is being tested is a 2.2Litre, 4 Cylinder turbo DW12BTDE4 engine made by Ford/JLR across a set number of engine speeds (rpm). The purpose is to record how the engine reacts in its torque, power output and thermal efficiency and then compare the results with other published data of other Turbocharged Diesel cycle and naturally aspirated Otto-cycle engines which will help strengthen the understanding in the relationship in the differences of each cycle and their use cases.

The use of a dynamometer allows the recording of key engine factors such as torque, speed and fuel flow under restricted conditions providing the necessary data to further determine key metrics such as the Brake power, BMEP, BSFC and thermal efficiency.

Experimental Setup

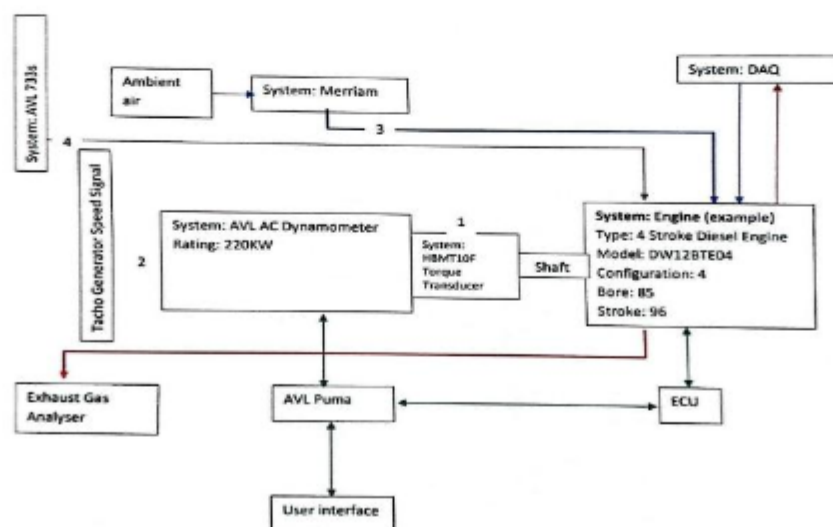


Figure 1: Schematic of the engine dynamometer setup

Equipment

Engine	
Engine Manufacturer	Ford/JLR
Engine Model	DW12BTED4
Aspiration	Turbocharged
Engine Displacement	2.2 Litre
Stroke Cycle	4
Bore	85mm
Stroke	96mm
Dynamometer	
Model	AC 220/250 kW AVL dynamometer
Sensors	
AVL733s gravimetric fuel meter	Weighs fuel consumption (kg/h)
AVL PUMA control system	Logs and controls load
Torque Flange	Measure's reaction torque
Merriam Mass airflow meter	Volumetric air intake sensor
Indicom in-cylinder pressure analysis	Pressure trace in cylinder 4 via glow plug port

Table 1: Engine Specification and Sensor list

Experimental Procedure

1. Make sure engine and dyno are attached to each other with a driveshaft
2. Connect all necessary sensors to their respective positions
3. Complete safety checks (engine is secure, door is shut, fumigation extraction is on, emergency stop is ready)
4. Allow dyno to start turning over the engine.
5. When dyno gets to 1000rpm, start recording data (engine speed, torque, fuel flow)
6. Increase engine speed up to 3500rpm and record data every 250rpm once engine speed stabilises

Data and Calculations

Data from Experiment

Engine Speed (rpm)	Torque (Nm)	Fuel Flow Rate (kg/h)
1000	181.84	5.272
1250	275.08	8.996
1500	365.47	13.217
1750	421.98	17.613
2000	409.20	18.574
2250	410.54	21.190
2500	403.37	23.475
2750	395.47	25.771
3000	385.08	27.943
3250	366.74	29.413
3500	349.68	31.069

Table 2: Measurements form dynamometer test

Calculations

$$\dot{W}_b = 2\pi NT_b \quad bsfc = \frac{\dot{m}_f}{\dot{W}_b}$$

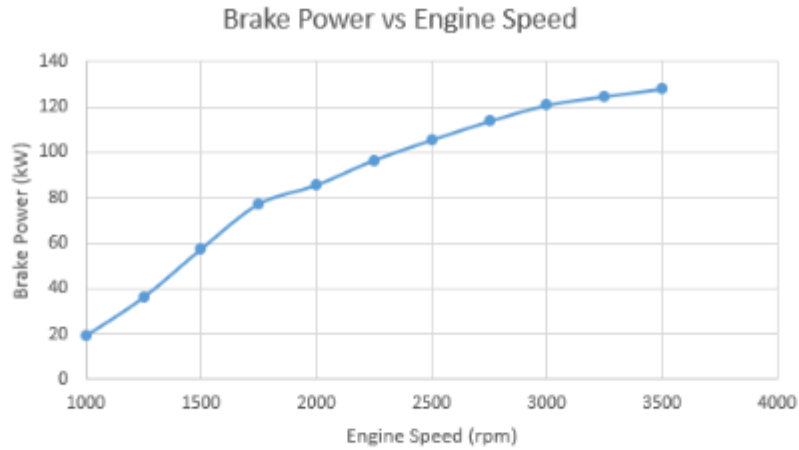
$$\eta_{th} = \frac{\dot{W}_b}{\dot{m}_f Q_{LHV}} \quad bmep = \frac{\dot{W}_b \cdot n_R}{V_d \cdot N}$$

Brake Power (kW)	BSFC (g/kWh)	BMEP (bar)	Brake Thermal Efficiency (%)
$\dot{W}_b = 2\pi NT_b$ $\dot{W}_b = 2\pi(1000)(181.84)$	$bsfc = \frac{\dot{m}_f}{\dot{W}_b}$ $bsfc = \frac{5.272(1000)}{(19.03206333)}$	$bmep = \frac{\dot{W}_b \cdot n_R}{V_d \cdot N}$ $bmep = \frac{(19.03206333)(2)}{(0.0022)(1000)}$	$\eta_{th} = \frac{\dot{W}_b}{\dot{m}_f Q_{LHV}}$ $\eta_{th} = \frac{(19.03206333)(3600)}{(5.272)(42600)} \cdot (100)$
$\dot{W}_b = 19.03206333$	$bsfc = 277.0062241$	$bmep = 10.48644752$	$\eta_{th} = 30.50727201$

Table 3: Example Calculations

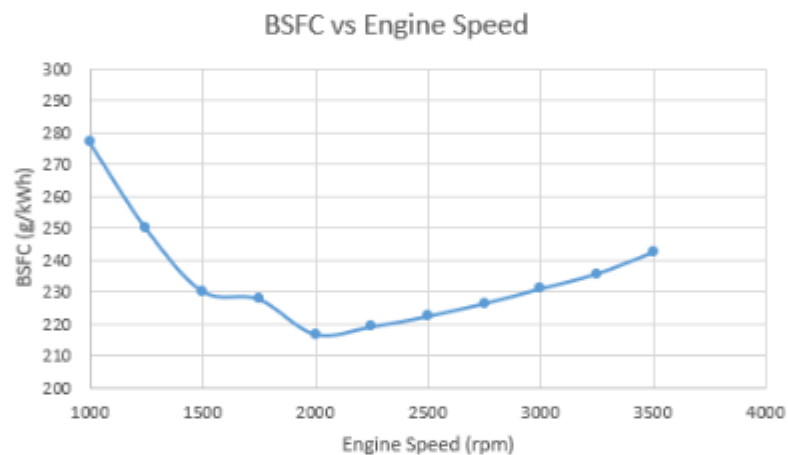
Engine Speed (rpm)	Brake Power (kW)	bsfc (g/kWh)	bmep (bar)	Brake Thermal Efficiency (%)
1000	19.03206333	277.0062241	10.48644752	30.50727201
1250	35.98963333	249.9608684	15.8638985	33.80810877
1500	57.379418	230.3439188	21.07693195	36.68733375
1750	77.29230367	227.8752109	24.3355248	37.08478948
2000	85.66003733	216.8339004	23.5988466	38.97316892
2250	96.6824055	219.1712121	23.67595156	38.55754661
2500	105.548745	222.4090869	23.26245675	37.99621833
2750	113.8297362	226.3995408	22.80686275	37.3265078
3000	120.916376	231.0935948	22.20784314	36.56831871
3250	124.7534037	235.7691184	21.15005767	35.84313452
3500	128.0976083	242.5416087	20.16580161	34.84228652

Table 4: Calculated results



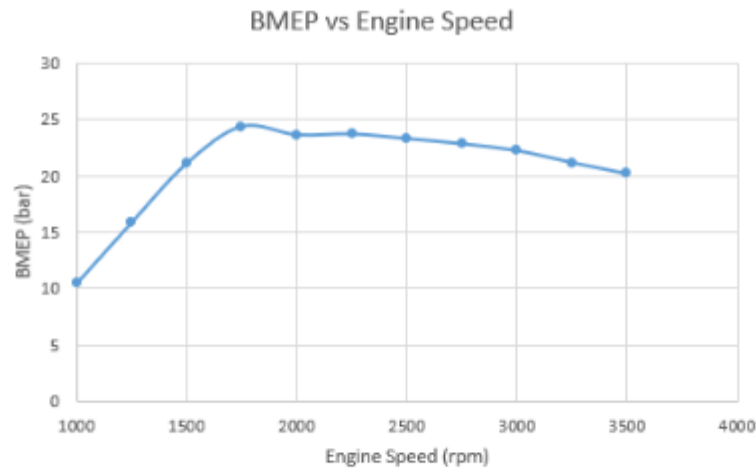
Graph 1: Brake Power vs Engine Speed

The brake power outputted in turn with the engine speed shows a linear increase up to 1750rpm which then increments at a shallower increase with the maximum power output of the engine recorded at around 128.1 kW. This is expected with diesel engines as torque drops off above the mid-range rpm.



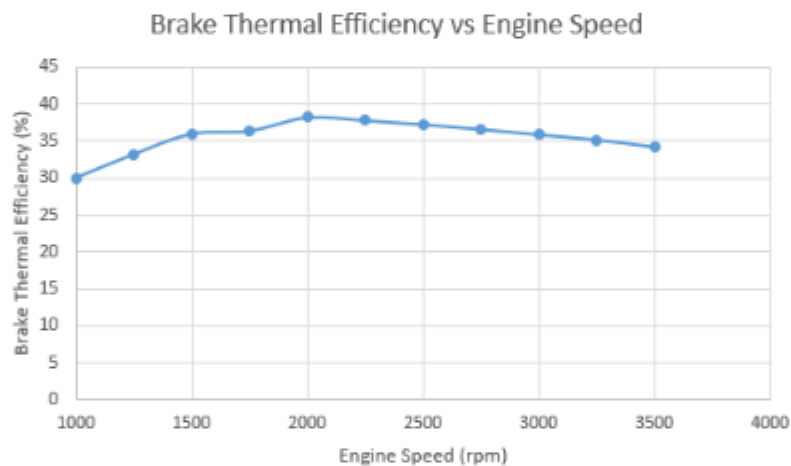
Graph 2: BSFC vs Engine Speed

In the BSFC we can further clarify the inefficiencies of diesel engines outside mid-range engine speeds as the minimal fuel consumption is at 2000 rpm at 218.83 g/kWh which is where the engine is at its most efficient fuel-air ratio as well as the lowest mechanical losses as where BSFC improves by decreasing the thermal efficiency will be at its peak [Graph 4].



Graph 3: BMEP vs Engine Speed

BMEP is at its maximum pressure at 1750rpm before declining due to the cylinder pressure being at its most effective operating condition as at higher speeds the increased friction of the piston against the engine block walls.



Graph 4: Brake Thermal Efficiency vs Engine Speed

This is inversely represented in the brake thermal efficiency which peaks again at 2000rpm at around 38.97% which highlights the most effective conversion of fuel-air mixture burnt into useful mechanical work. The data points towards the Ford/JLR DW12BTED4 being most efficient at its medium engine speeds with the fuel economy, thermal efficiency and torque output diminishing at lower and higher rpms due to incomplete combustion and increasing mechanical losses which is expected for turbo diesel engines.

Lab Test	Engine Speed (rpm)	Brake Power (kW)	bte (g/kWh)	bmp (bar)	Brake Thermal Efficiency (%)	
1000	15.0226303	277.9062341	11.48544792	39.39727021		
1250	35.9595333	245.9608904	15.2639585	33.8912977		
1500	57.379438	236.3409168	21.07950295	35.6970375		
1750	77.7829367	227.8751168	24.5555748	37.0479348		
2000	85.6603753	216.8359964	25.568466	38.97918852		
2250	96.681055	215.1717171	25.67955796	38.55754961		
2500	105.548745	222.6909868	25.26346673	37.9451833		
2750	113.8297362	226.3055608	25.80446775	37.5245078		
3000	120.316576	231.2655348	25.2674514	35.5651871		
3250	124.7551857	235.7691184	25.84515152		Displacement: 0.00175	
3500	128.0770807	242.3439977	26.3500201	34.8422892	Q_{br}	43001

Mondeo	Engine Speed (rpm)	Brake Power (kW)	bte (g/kWh)	bmp (bar)	Brake Thermal Efficiency (%)	
997.75	15.7908963	255.730312	9.48270432	31.6327303		
1229.75	28.3030775	257.1333432	13.58114437	32.2705700		
1483	44.1063689	235.7117471	17.7295997	35.6129632		
1805	61.6476811	225.8254806	20.63610448	37.8904934		
2097.5	69.1872765	223.9648902	20.7982015	37.0928508		
2342.5	76.5268289	222.1709301	20.4640003	37.9620629		
2499.25	82.3572130	222.049331	19.84053796	38.1262111		
2696	90.5866988	226.3168597	18.294908	37.39634712		
3360.75	96.58124834	240.2867938	16.66457227	35.31254978		
3995.25	98.83348604	247.1285965	13.63108993	34.17448176		
4195.75	88.9810781	253.3278904	12.4594055	33.35402176	Displacement: 0.002	
4495.75	79.25416452	265.5896823	10.50813648	31.89000216	Q_{br}	42001

GM	Engine Speed (rpm)	Brake Power (kW)	bte (g/kWh)	bmp (bar)	Brake Thermal Efficiency (%)	
1200	21.1035705	289.157055	8.01726244	28.6493992		
1600	72.2901804	281.632755	9.03642090	29.0388221		
2000	94.1579185	272.844241	9.43375330	30.4017079		
2500.0	136.808802	261.827184	9.71775925	31.6684295		
2800	156.088769	260.476065	9.79934367	31.8293738		
3200	158.751736	265.6429	9.627142101	31.2293795		
3600	180.793629	269.488118	10.1520355	30.7894738		
4000.2	206.5016572	263.483176	11.51048186	29.26281641		
4400.1	224.3832201	291.43645	10.18867738	28.4536597		
4800.2	258.350342	305.769693	9.038178915	27.12610794		
5000	242.3075234	312.236003	9.03030016	26.4011503		
5199.9	240.5995440	319.109761	9.20830890	26.3670388	Displacement: 0.006	
5400	257.052807	329.481311	9.76205830	25.1054045	Q_{br}	43001

CAI	Engine Speed (rpm)	Brake Power (kW)	bte (g/kWh)	bmp (bar)	Brake Thermal Efficiency (%)	
1397	30.18	212.7	20.29	29.4039633		
1200	329.6	212.7	20.47	26.4733989		
1391	422	212.7	20.55	29.2824526		
1700	425.3	212.7	18.8	49.94281179		
2000	409.2	212.7	15.32	38.6279335	Displacement: 0.016	
2100	387.7	213	14.2	48.6576232	Q_{br}	42001

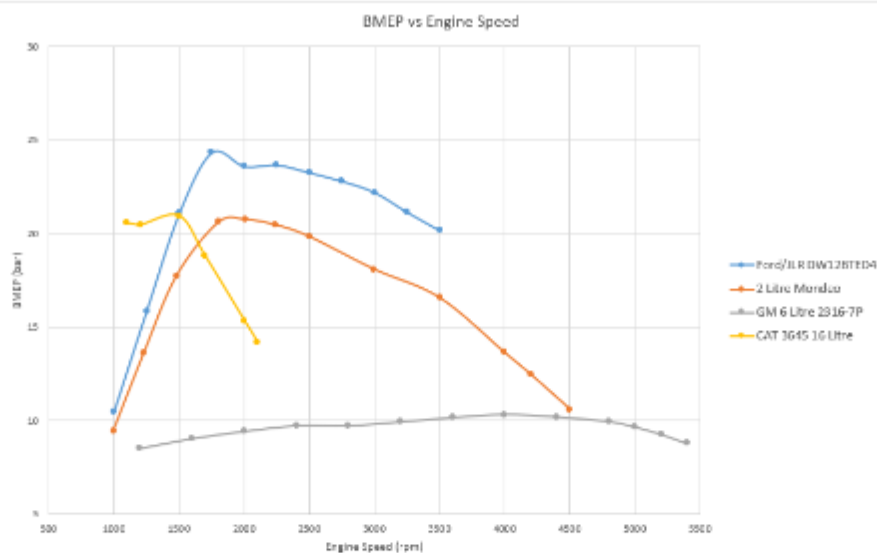
Figure 1: Data tables of all engines

The data above shows that the Lab (DW12BTED4) and Mondeo which are turbo diesel engines are directly comparable to each other with the only difference is the engine displacement by being more modern and optimised having a peak efficiency over a larger range of engine speeds with the lab tested engine having a narrower sweet spot.

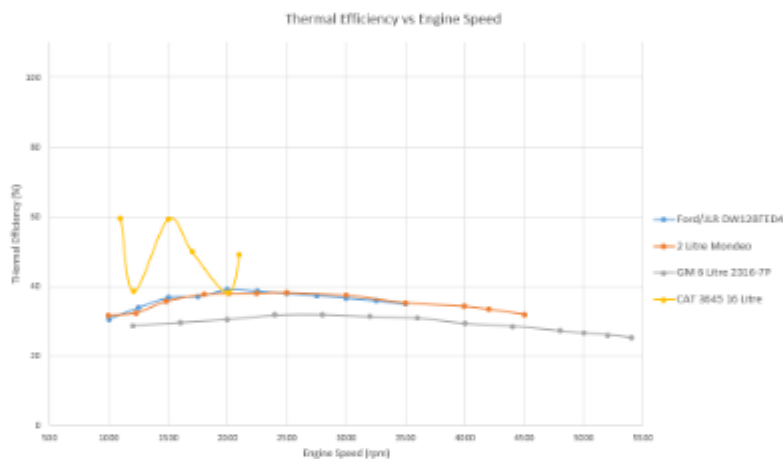
The GM which is Naturally Aspirated (Otto) is the worst performing engine in all aspects and needs to operate at higher RPMs to develop useful power and not as fuel efficient than the turbo charged diesel engines which highlights the inherent efficiency disadvantages of the otto cycle in terms of compression ratio and throttle losses. [1]



Graph 5: BSFC vs Engine Speed



Graph 6: BMEP vs Engine Speed



Graph 7: Brake Thermal Efficiency vs Engine Speed

Comparing the data from all the engines above, in terms of BSFC (Graph 5), the overall trend of the U shaped trough with the JLR and Mondeo engine's fuel consumption significantly worsen after their optimum engine speeds, the Mondeo has the most efficient fuel consumption (225.855-222.049 g/kWh) from 1805 to 2499.25rpm with the CAT having an atypical constant fuel efficiency of 272.7g/kWh due to not being tested at full open throttle. With the BMEP (Graph 6), the diesel engines peak sharply in their cylinder pressure at lower speeds (1500-1800rpm) and then decline with the gasoline engines peak being far later and broader. The Mondeo's pressure is significantly reduced as it reaches its proper redline of 4500rpm which reinforces the zero-performance benefit of over revving a diesel engine past its torque peak. The GM peaks at 10.32 bar at 4000rpm where the Mondeo is hitting at its maximum rpm, showing petrol engines enter their optimal power zones when diesel engines have finished.

The thermal efficiency (Graph 7) is almost an inverse of the BSFC values with the efficiency rising in the mid speed range where the BSFC is at its lowest and BMEP is at its highest. The CAT engine having the highest efficiency of 59% dropping to 31.87% as the rpm increases. GM's efficiency is consistently lower but across a wider range of speeds. The Mondeo and Lab

engine having near identical efficiency due to the engines both being 4 cylinders but the Mondeo having a smaller displacement at 2 Litres as well as having data all the way to its redline whereas the lab engine only has up to 3500rpm.

Conclusion

This report has demonstrated the trade-offs between different engine types due to the thermodynamic properties of both the Otto and Diesel cycle and their importance in combustion ratio and engine management with Turbo diesel engines dominating in low-speed torque and higher thermal efficiency making them ideal for pulling applications and fuel economy with the naturally aspirated petrol engine, sacrificing its efficiency and low-end torque for a broader range of power and higher revving capacity. The brake power increased in turn with the engine speed with torque and BMEP producing their highest values at 1750-2000rpm along with the lowest BSFC and highest thermal efficiency around those speeds as well highlighting that the median engine speeds provide the best productive output of the DW12BTED4 engine. In comparison with a naturally aspirated petrol engine, it produced a stronger torque curve along with higher fuel efficiency.

Reference

[1] Heywood, J. B. (2018). *Internal Combustion Engine Fundamentals*.



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Wind Tunnel

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Wind Tunnel Lab Report

Abstract

This report investigates the aerodynamic drag and lifts on four basic shapes in a closed return wind tunnel at set speeds to determine their coefficients. A theoretical analysis on the development of the boundary layer on the ground board was also undertaken, calculating the Reynold's number, layer thickness and friction coefficient. These results confirm a transition in flow from laminar to turbulent between 3 and 6 m s^{-1} marked by the shift in friction coefficient. The experiment demonstrates the direct relationship between an object's geometry, flow regime and its resulting aerodynamic forces linking empirical data with fundamental fluid dynamic principles.

Introduction

The purpose of a wind tunnel is to simulate airflow over solid objects in a controlled, measurable and repeatable environment to circumvent the difficulties and expenses of real-world testing. It enables engineers to study aerodynamic drag and lift forces. Theoretically, the boundary layer thickness will be calculated for a set range of windspeeds, practically in the experiment, a cube at 0 and 45 degrees of incidence to the wind are being tested alongside a rectangular cuboid and a streamline body (teardrop shape) to which with the windspeed, air density and forces. The Drag and Lift Coefficients can be calculated for each shape and compared with other examples to validate the accuracy of the results whilst also noting the errors that have occurred.

Experimental Setup

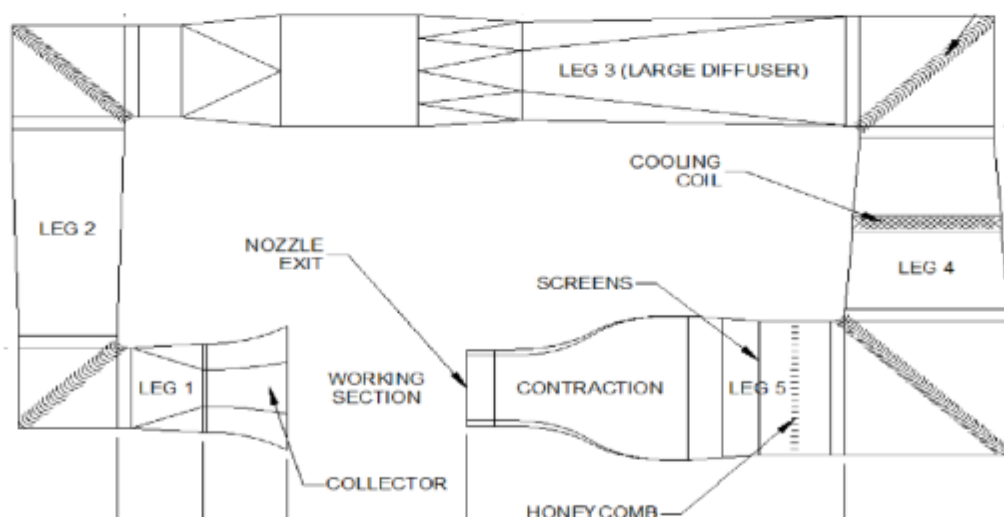


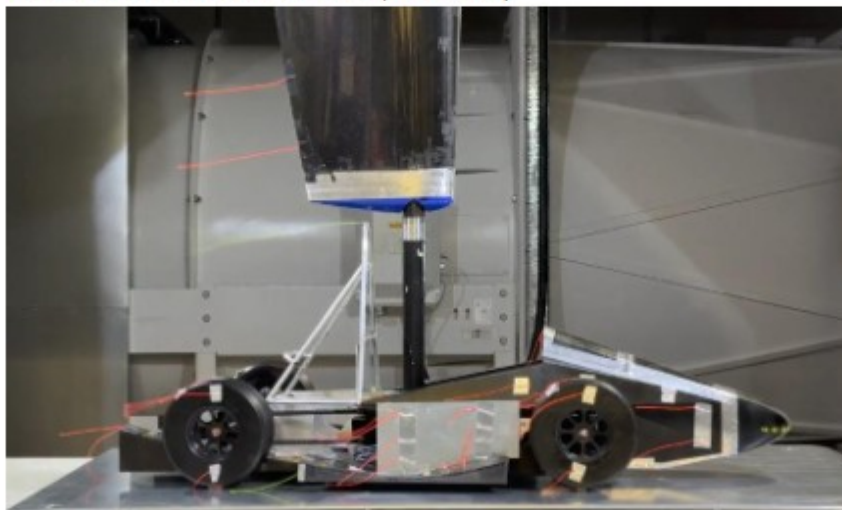
Figure 1: Schematic of Coventry University Wind Tunnel

Wind Tunnel Information

Wind Tunnel	
Type	Closed single return tunnel with open jet working section
Comer turning vanes	Rolled steel vanes bonded into assemblies
Air settling	Three stainless steel setting screens 500 mm apart at the entrance to the contraction
Mesh screen specification	57.4% open area, 1 mm aperture size (2 off 2460 mm x 1890 mm, 1 off 2286 mm x 1890 mm)
Air straightening	Honeycomb wall in settling chamber. 75 mm thick x 5.2 mm core size (1 off 2286 mm x 1890 mm)
Temperature control	Water cooled to $\pm 2^{\circ}\text{C}$ by a cooling coil
Working Section	
Type	Open Jet
Length	2.5m
Contraction Ratio	3.63:1
Maximum Air Velocity	45.0 m/s
Nozzle Width	1300 mm
Nozzle Height	1000 mm
Fan Section	
Fan Power	110kW (147.5 hp) variable speed DC motor
Data Acquisition and Control	
Model Balance	Bespoke design by Mercedes AMG F1

Methodology

1. Null sensors so they are all set to 0
2. Mount model to the strut inside the working section and add tufts for airflow visualisation onto the model (as shown)



3. Run the wind tunnel and record the values at the 10,20,30ms⁻¹ mark
4. Turn off wind tunnel and wait for it to stop before changing the model
5. Change carefully
6. Repeat 1-6 until all models have been tested

Data and Calculations

$$Re_L = \frac{\rho V L}{\mu}$$

Laminar: $\delta \approx 5.0x/Re_x^{1/2}$
 $c_f = 0.664/Re_x^{1/2}$

Turbulent: $\delta \approx 0.16x/Re_x^{1/7}$
 $c_f = 0.027 / Re_x^{1/7}$

$$F_D = C_D A \frac{1}{2} \rho V^2$$

$$F_L = C_L A \frac{1}{2} \rho V^2$$

$$\text{Dynamic Pressure (q)} = \frac{1}{2} \rho V^2$$

$$C_D = F_D / Aq$$

$$C_L = F_L / Aq$$

Theoretical Calculations

Assuming $\rho = 1.225 \text{ kg/m}^3$ and $\mu = 1.7894 \times 10^{-5} \text{ kg/(ms}^{-1}\text{)}$

Reynolds Number (Re_L)	Friction Coefficient (C_f)	Boundary Layer Thickness (m)
$Re_L = \frac{\rho V L}{\mu}$ $Re_L = \frac{(1.225)(1)(1.29)}{(1.7894 \times 10^{-5})}$	$c_f = \frac{0.664}{Re_x^{1/2}}$ $c_f = \frac{0.664}{88,311.7246^{1/2}}$	$\delta = \frac{5.0x}{Re_x^{1/2}}$ $\delta = \frac{5.0(1.29)}{88,311.7246^{1/2}}$
$Re_L = 88,311.7246$	$c_f = 0.00223439$	$\delta = 0.021704537285m$

Table 1: Example Calculations for Reynolds number, Friction Coefficient and Boundary Layer Thickness

Velocity (ms^{-1})	Reynolds number (Re)	Laminar (L) Or Turbulent (T)	Boundary Layer Thickness (m)	Friction Coefficient (c_f)
1	88,311.7246	L	0.021704537285	0.00223439
3	264,935.1738	L	0.012531120444	0.001290025
6	529,870.3476	T	0.031403003	0.004107951
10	883,117.2460	T	0.029192985	0.00381885
20	1,766,234.4920	T	0.026440777	0.003458823
30	2,649,351.7380	T	0.024952744	0.003264167
40	3,532,468.9840	T	0.023948037	0.003132737

Table 2: Theoretical Results for Reynolds number and Boundary layer thickness

Theoretical Graphs

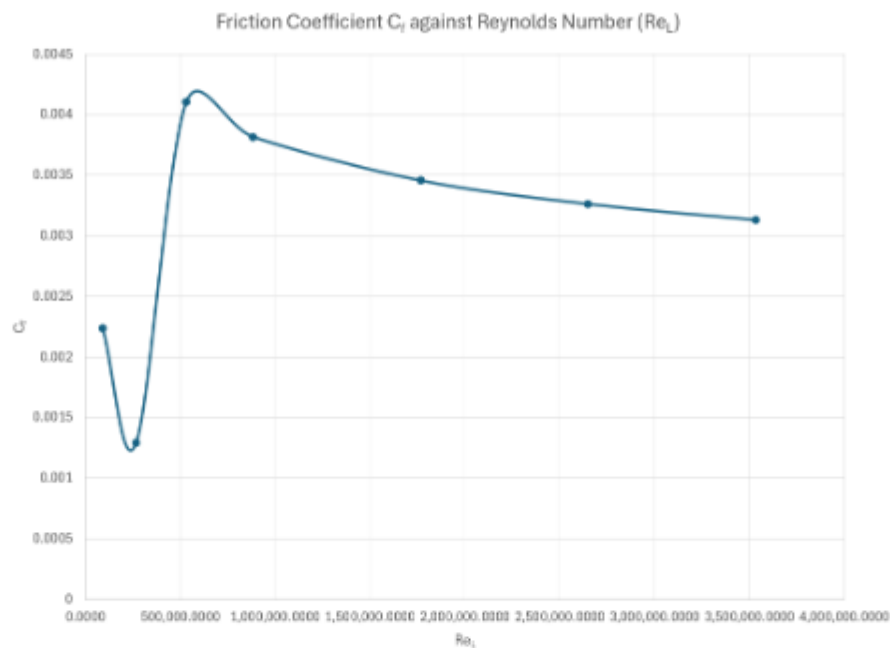


Figure 1: Friction Coefficient against Reynolds Number

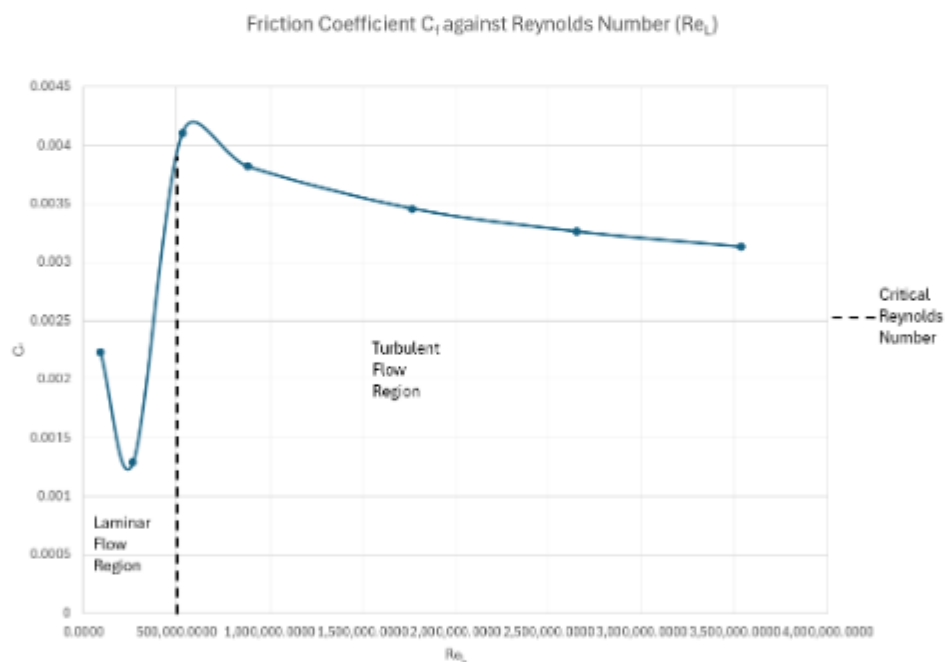


Figure 2: Friction Coefficient against Reynolds Number with $Re_{critical}$ marked dividing the laminar and turbulent flows.

The theoretical data above illustrates the relationship between the skin friction coefficient (C_f) and the Reynolds number (Re_L) showing a behavioural shift between the laminar and turbulent flow regimes. The overall trend is a monotonic decrease in C_f as Re_L increases in the turbulent flow region. This is because at higher Re_L 's there is a greater dominance of inertial forces over viscous forces leading to a thinner boundary layer and reduced average shear stress on the surface [1]. In the laminar region, C_f decreases sharply conforming to the

Blasius solution [2]. The transition to turbulent flow is marked by the increase in the magnitude of C_f . In the turbulent flow region continues to decrease at a shallower gradient follow the decay characteristic of the Re_L increasing [3].

Results from Experiment

Model	Frontal Area (m ²)	Wind speed (ms ⁻¹)	ρ (kg/m ³)	Dynamic Pressure (q)	F_D (N)	C_D	$C_{D, \text{average}}$	F_L (N)	C_L
Cube (0°)	0.02235	10.030619620156	1.14285139774	57.4930423905747	1.643383412123	1.2789279554188		-0.0541594517467343	-0.042148433760555
		19.9662718989678	1.14764112621	228.754722893646	6.481137846794	1.26766255484896	1.26686896710963	-0.220798103350258	-0.0431864734621437
		29.9695008442297	1.14522061018	514.301959366607	14.41447796676	1.25401639126112		-0.698499556525527	-0.0607673682923802
Cube (45°)	0.0322	9.97028407975325	1.14356831704	56.8389995023067	1.7084390886210	0.93346320951801		-0.137029002101462	-0.0748704077963456
		19.934324008363	1.14236900628	226.975740821436	6.9306583231920	0.94828564282679	0.94205768676296	-0.51816772469527	-0.0708681731940141
		29.9295903861785	1.13941955749	510.334842482438	15.519517054950	0.94442420784309		-1.1100214938587	-0.0675492134397372
3D Rectangle	0.02235	10.0185322498414	1.1422646769	57.3251173407322	1.293400456926	1.00950977884857		-0.078303065130277	-0.0611161914622522
		19.9515959844267	1.14166023442	227.228165513943	5.181778542828	1.02032648172613	1.01256625722234	-0.240274415729398	-0.0473116222979643
		30.0288002396566	1.13925168407	513.579633511808	11.56875468037	1.00786251109231		-0.538441199452415	-0.0469086530356077
Streamline Body	0.02235	9.93182120798216	1.14279177301	56.3608330934013	0.41516916534	0.32958706537242		0.03688895227735442	0.0292851940099417
		19.9829843284339	1.14243413835	228.096207374621	1.5533807255730	0.30470425048131	0.30843440529763	-0.0596784239398072	-0.0117062540670966
		30.0613035144136	1.13955095353	514.895824724445	3.3489421543530	0.29101190003916		-0.200791391582254	-0.0174481020222793

Table 2: Results from Wind Tunnel test

Example Equation

$$q = \frac{1}{2} (1.142791773)(9.93161208)^2$$

$$= 56.36083$$

$$C_D = (0.415169165)/(0.02235)(56.36083)$$

$$= 0.329587065$$

Experimental Data Graphs

Lift & Drag Forces

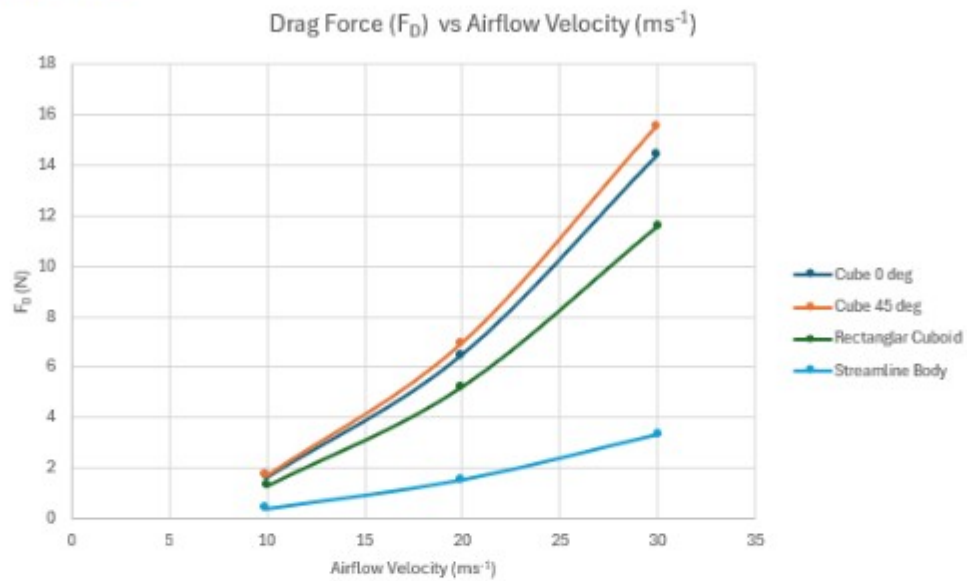


Figure : F_D vs Airflow Velocity

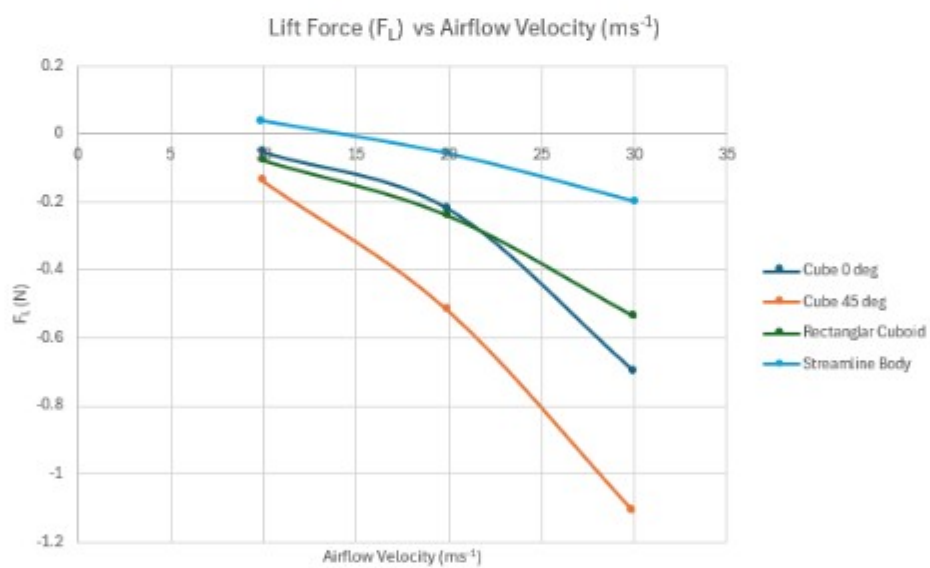


Figure : F_L vs Airflow Velocity

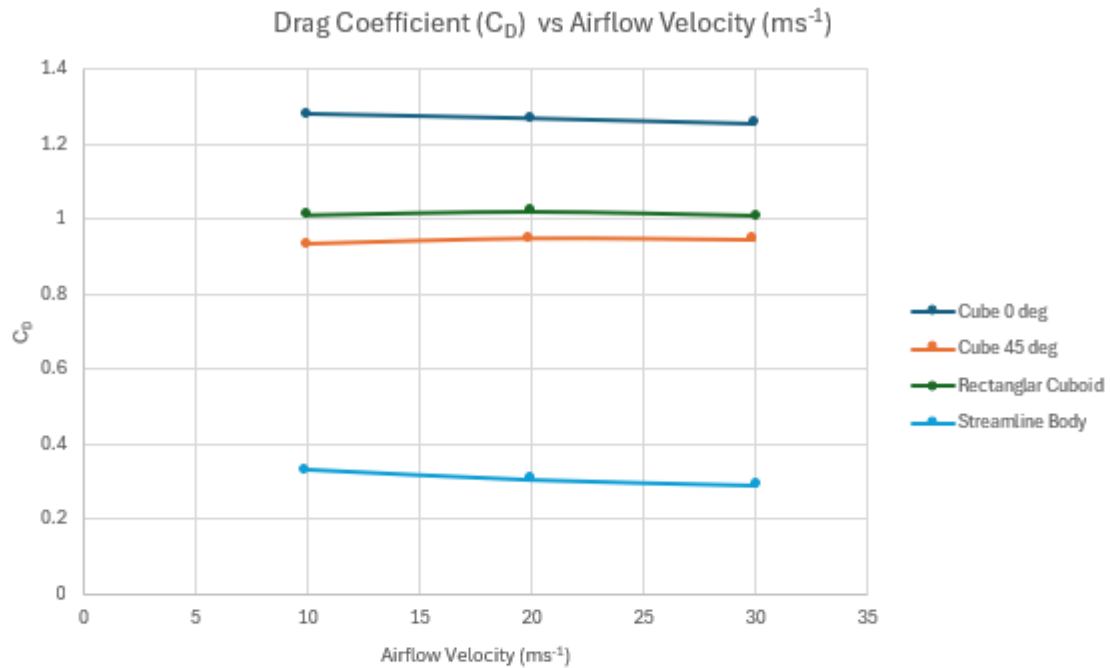


Figure : C_D vs Airflow Velocity

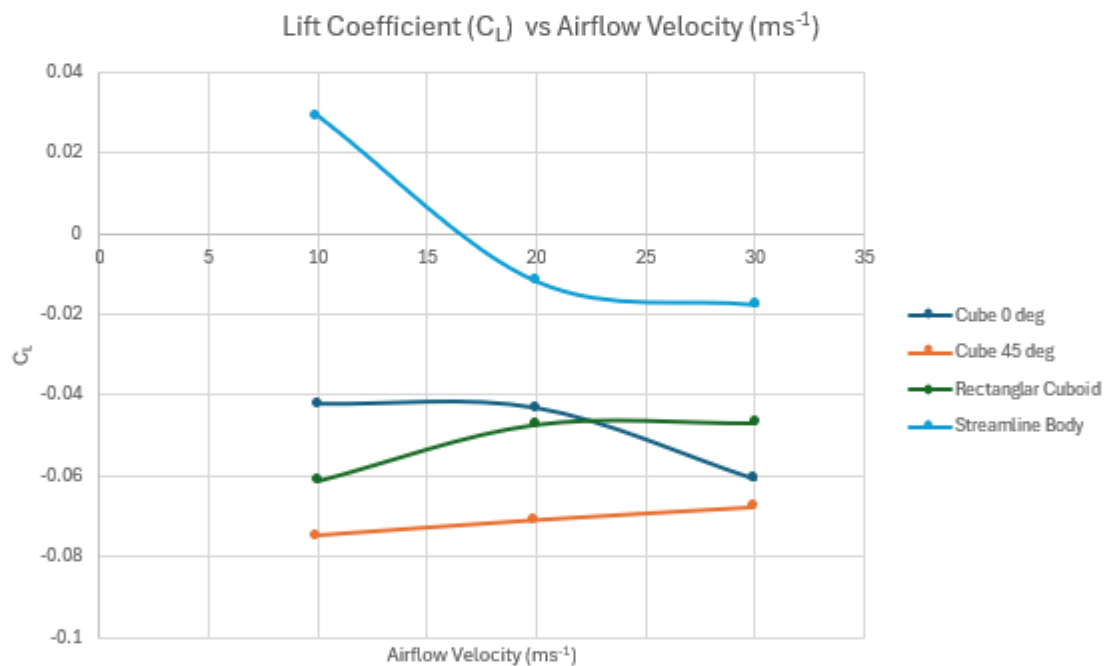


Figure : C_L vs Airflow Velocity

Comparison with other experimental observations

With the four basic shapes they can now be compared to other known sources to validate the data results obtained during the testing period, including the systematic errors in the results, with the Cube at 0° of incidence having a $C_{D, \text{average}}$ of 1.267 is slightly higher than other recorded values as it should be around 1.05 [4]. This error is also present when the Cube when mounted at 45° with the $C_{D, \text{average}}$ of 0.942 which is higher than the reported value of 0.80 [4].

The rectangular cuboid ($C_{D, \text{average}} = 1.013$) sits closer with published data as even the increase in streamwise length can see a reduction in the drag coefficients as the flow has a higher chance of reattaching to the body reducing the width in the wake due to the aspect ratio of the shape being greater [4].

The streamline body's the results obtained are significantly higher ($C_{D, \text{average}} = 0.308$) than the ideal C_D of less than 0.1 which is down to model specific imperfections such as the surface roughness, support structure interference which will affect the pressure stabilisation at the rear of the shape. Nonetheless the streamline body is still experiencing around 71.4% less drag than the bluff bodies of the cube orientations and the rectangular cuboid.

However, the results obtained from the experiment are littered with interconnected sources of aerodynamic interference which affect the results which are theoretically supposed to be in free-stream conditions. The primary sources of these errors stem from the support strut due to its size generating upstream turbulence and a disruptive wake which alters the pressure and velocity distribution around the top and rear of each object. Wind tunnel blockage could also be playing a factor in the lack of free flow is also exacerbated by the ground effect where the proximity of the ground board and its boundary layer decreases the pressure and accelerates the flow under the object and creates a non-uniform velocity profile. These interferences explain the generation of lift. For a symmetrical object in an ideal flow, the lift coefficient should be zero however this is not the case and is an experimental artifact arising from the disruption of the airflow.

Conclusion

The theoretical calculations in this report has shown the theoretical data showing the transition phase of laminar to turbulent flow with the change in the friction coefficient as Reynolds Number increases highlighting the impact of the boundary layer thickness in wind tunnels and real-world applications. Experimentally, the variation between the cube mount and their differences in aerodynamic properties in terms of the lift and drag experienced on each shape. With Experimentally, With the larger frontal surface area giving higher drag coefficients and disruptive airflow with the higher aspect ratio bluff bodies having a more attached flow to their lower aspect ratio counterparts which confirm stable drag coefficients across windspeeds of 9ms^{-1} to 30ms^{-1} with the systematic errors in the wind tunnel testing phase clearly outlined and explained proving it is a fundamental issue with the setup and not human error in any way. The cube at 45 degrees of incidence having the lowest average drag of the bluff bodies due to its decreased surface area and higher attached flow with the rectangular cuboid having the advantage of the elongated aspect ratio allowing flow to reattach further down the shape itself with the cube having highest C_D as flow fails to reattach to the shape.

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